Highlights

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ABSTRACT

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1. Introduction and literature review

Analysing public transport supply from observed operation can be a very complicated task for governmental entities. The quantities that determine whether capacity is enough, such as peak loads and the occurrence of full buses, are extreme values, and estimating them reliably requires a large number of observations taken under the same conditions [1]. The binding constraint, however, is not the volume of data itself but the length of the period over which the service configuration remains unchanged. Each modification of the supply thus restarts the sample. For this problem, multiple entities rely on simulation techniques to analyse supply without real deployment [2].

This work focuses on the capacity of buses to absorb demand studying a digital simulation of a bus route in Madrid, Spain, using a multi-agent system.

The multi-agent paradigm is a well-known and widely implemented technique in transport simulation, and its use has increased in recent years [3–5]. Despite this, in public transport, the main area of study in which it applies is mostly real-time control systems [6], whereas this work focuses on demand level analysis.

Some of the works among the literature of MAS in transportation take advantage of the cognitive contribution certain paradigms such as BDI (Belief-Desire-Intention) offer, like modeling drivers and evacuees who must take decisions during a simulation [7, 8]. Such architectures are appropriate when the decision process of each individual agent is the object of study; when the focus is the aggregate behaviour of the system, simpler agents are enough [9]. The architectural decision of which types of agents to use for a certain MAS depends on the purpose the simulation aims at and on the set of data available.

In terms of granularity and nature of the simulation, different types appear in literature: macroscopic, mesoscopic and microscopic [10]. Microscopic simulations refer to those which describe the individual exact behaviour of entities, with a level of detail up to second by second. On the other hand, macroscopic simulations do not represent that individual behaviour, but rather use a global perspective to obtain aggregated level indicators and without the need to model invidividual entities. In the middle of those appear the mesoscopic. While it is used to represent aggregated level indicators, it still relies on using individual entities as so, but without the need of having them modelled at detail [2].

The mesoscopic approach seems to suit better for the purpose of this work, as each bus is represented as an entity, but its movement among the route comes from sampled data from fitted distributions and does not change by traffic dynamic inside the simulation. A microscopic level of detail is something this kind of problem does not require [2].

In this work passengers are used to analyse bus capacity, and the origin and destination of individual entities are not required. No external surveys or other data from which to construct OD matrices are available. The usual approach is to infer the matrices via IPF with null seeding [11, 12], while in this work the boarding and alighting of passengers is done via fitted distributions from real data operations. This decision comes from the goal of the work. OD matrices assigns origin and destination to passengers, but here the problem requires only to know how many boards and alightings occurs in a specific stop given a bus arrival. Several bus simulators already operate from stop-level counts without an

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OD matrix. BusMezzo [2] is a mesoscopic simulator in which passengers are not represented individually. Arrivals at stops follow a Poisson process, and the number of alighting passengers is drawn from those on board. BusSim [13] is an agent-based model with bus and stop agents in which passengers represents aggregate quantities, built on synthetic data for short-term prediction. Talusan et al. [14] constructs an event-based simulator with APC (Automatic Passenger Counting) data to evaluate dispatching decisions. These works establish that occupancy can be studied without passenger destinations. However, first and third works are not multi-agent systems, the second is not calibrated on real operation, and none of them models the same line with and without individual passengers.

For the agents used in this paper, as the work studies no individual decision and the available data contain no information about one, no cognitive reasoning is represented for either the bus or the passenger. The agents are therefore reactive in the classification of Wooldridge and Jennings [15]: they act in response to events and to the messages they receive. They are not purely reactive, however, as they keep an internal state [16]. The form they have of interaction is through FIPA interaction protocols [17]. The communication between different entities is the core of the behaviour of the system.

From the engineering side, this work provides a multi-agent model of line E01 of EMT Madrid, calibrated with the boardings and alightings recorded by the operator and used to analyse occupancy and demand absorption. From the artificial intelligence side, it addresses how fine the agents of such a system need to be. Two solutions are built that share the same bus model and similar interaction protocols between reactive agents, and differ only in whether each passenger is an agent or passengers are counts held by stops and buses. Comparing them isolates what the passenger agent contributes, in results and in computational cost, when the available data record only counts.

The remainder of the paper is organised as follows. Section 2 presents the two proposed solutions, defining the agents and the interaction protocols between them. Section 3 describes the operational data and how the boarding and alighting distributions are calibrated from them. Section 4 details the implementation in GAMA, the simulation parameters for line E01 and the results obtained. Finally, Section 5 draws the conclusions.

2. Proposed solutions: two different approaches

Transport simulations usually represent each passenger as an agent, because a passenger is an entity that makes choices [18]. The data available here record only how many passengers board and alight at each stop, so there is no individual choice to represent. Two solutions are therefore built. The first solution keeps the passenger as an agent, as far as the data allow, but the decision to alight is not its own. As a bus approaches a stop, the environment draws the number of alighting passengers from the fitted distribution and marks that many passengers on board, which then request to alight when the bus asks. The second solution removes the passenger agent, and buses and stops keep only the number of passengers. Comparing the two shows what the passenger agent contributes when its behaviour cannot be derived from the data.

2.1. Agents and protocols

As introduced in Section 1, the agents modelled in this work are reactive in the classification of Wooldridge and Jennings [15]. Their behaviour is a fixed set of rules activated by events and by the messages they receive, with no deliberation. They are, more specifically, agents with state [16], since each one keeps information between events, such as the occupancy of a bus or the queue of passengers waiting at a stop, and acts according to it. Each agent controls its own state and decides its response to a request according to it; a bus, for instance, accepts or refuses a boarding request depending on its remaining capacity. This section defines the agents of the two solutions and the interaction protocols through which they communicate.

In the two proposed implementations, three agents are defined:

- Bus: Representing a phisical bus inside the simulation. It has a list of the trips it is scheduled to run, and keeps track of which and how many passengers has on board. It can present three different modes, which relates to its *scheduled starting point*, the *communications when its at a stop*, and the *transit between stops*.

- Stop: Representing each stop of the bus route. It keeps track of which and how many passengers are waiting to the bus in what its considered its *passengers custody* mode. When a bus arrives, it changes to *communicating with a bus*.

Table 1
Interaction protocols of the two solutions, in order of execution during a stop visit.

Protocol	Initiator → participant	Type	Purpose
First solution			
Alighting query	Bus → passenger on board	Query	Whether the passenger alights at the next stop
Alighting request	Bus → alighting passenger	Request	The passenger leaves the bus and is removed
Arrival notification	Bus → stop	Request	The stop starts notifying its waiting passengers
Boarding notice	Stop → waiting passenger	Request	Tells the passenger which bus to address
Boarding request	Passenger → bus	Request	The bus agrees, or refuses when full
Count update	Bus → stop	Request	Who boarded and their waiting times
Second solution			
Arrival query	Bus → stop	Query	Number waiting and number alighting
Count update	Bus → stop	Request	Number of passengers that boarded

- Passenger: This agent only appears in the first implementation. It represents each passenger individually, and has two possible modes:*waiting at a stop* and *on a bus*. It has a moment of creation timestamp, which represents at which exact time it appears at a stop.

The agents interact through FIPA protocols, specifically Request [19] and Query [20] types. Table 1 lists them in the order in which they run each time a bus serves a stop. Since the agents share a common vocabulary and their responses follow fixed rules, only the message paths that can occur in the simulation are implemented: a request is always agreed to, except in the boarding request.

In the first solution, as a bus approaches a stop it sends an *alighting query* to each passenger on board, which answers whether it will alight according to the mark set by the environment. The bus then sends an *alighting request* to each passenger that answered affirmatively; the passenger confirms and is removed from the simulation, and the bus updates its list of passengers, which frees capacity. Next, the bus sends an *arrival notification* to the stop, and the stop sends a *boarding notice* to each waiting passenger, containing the reference of the bus to address. Each passenger then sends a *boarding request* to the bus, including the time it has waited. This is the only protocol with a refusal: the bus agrees while it has capacity and refuses once it is full, in which case the passenger remains at the stop. Finally, the bus sends a *count update* to the stop with the passengers that boarded and their waiting times, which the stop records.

In the second solution there are no passenger agents and two protocols remain. On arrival, the bus sends an *arrival query* to the stop, which returns the number of passengers waiting and the number that must alight, the latter obtained from the environment. The bus applies both figures against its capacity and closes the visit with a *count update* carrying the number of passengers that boarded.

Every exchange consists of three messages, except a refused boarding request, which takes two. In the first solution the number of messages per stop visit therefore grows with the passengers on board and waiting, whereas in the second it is constant at six.

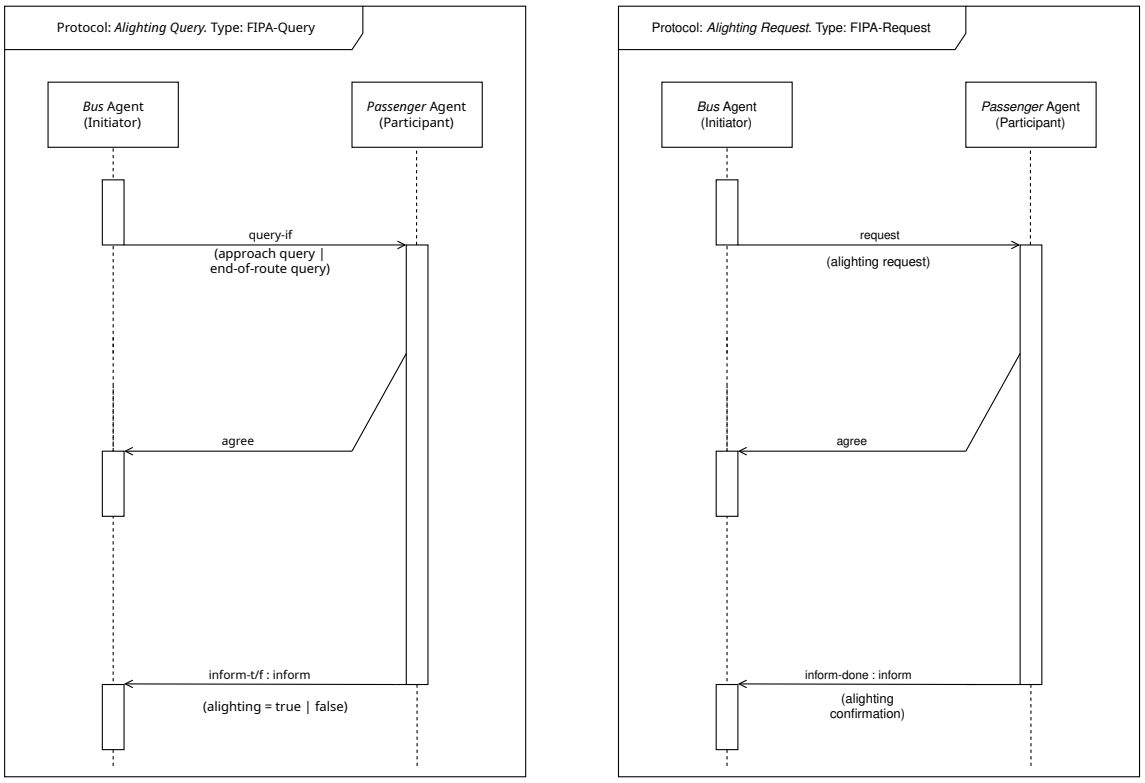
The FIPA diagrams that formalize this protocols are shown in Figures **??**, **??**,

2.2. Environment

Some elements of the model do not belong to the domain of any agent and are handled by the environment. Those are the passage of time, the movement of buses between stops, the generation of passengers, the selection of the passengers that alight, and the time a bus remains at a stop. They are the same in both solutions, except where stated.

Time and movement Time advances in discrete steps of length Δt, from the start of the service day until the last bus has completed all its scheduled trips. The movement of a bus between two stops ignores spatial position and is purely temporal. This process consists of a travel time to be completed and a counter that increases by Δt at each step. This avoids spatial computation and reduces the computational load.

Generation of agents Bus and stop agents are created before the simulation starts: one stop agent for each stop of each direction of the route, and one bus agent for each physical vehicle, which receives the list of its scheduled departures and the stop where its service begins. A bus agent is removed once it has completed its service.



(a) Alighting query (b) Alighting request

Figure 1: FIPA-Request interaction protocols from first solution (1/3)

Passengers are generated during the simulation. At the start of each hour h, a value τ_h is drawn for every stop from the boarding distribution of that stop and time slot. Since that distribution is fitted to the boardings aggregated over one hour, τ_h is the number of passengers that appear at the stop during that hour. They are created progressively. At each step an increment is added to a counter kept for the stop, and whenever its integer part reaches one or more, that number of passengers is created and subtracted from the counter. The increment is not constant within the hour but follows a line that pivots on the midpoint of the hour and reaches, at its end, the increment of the following hour:

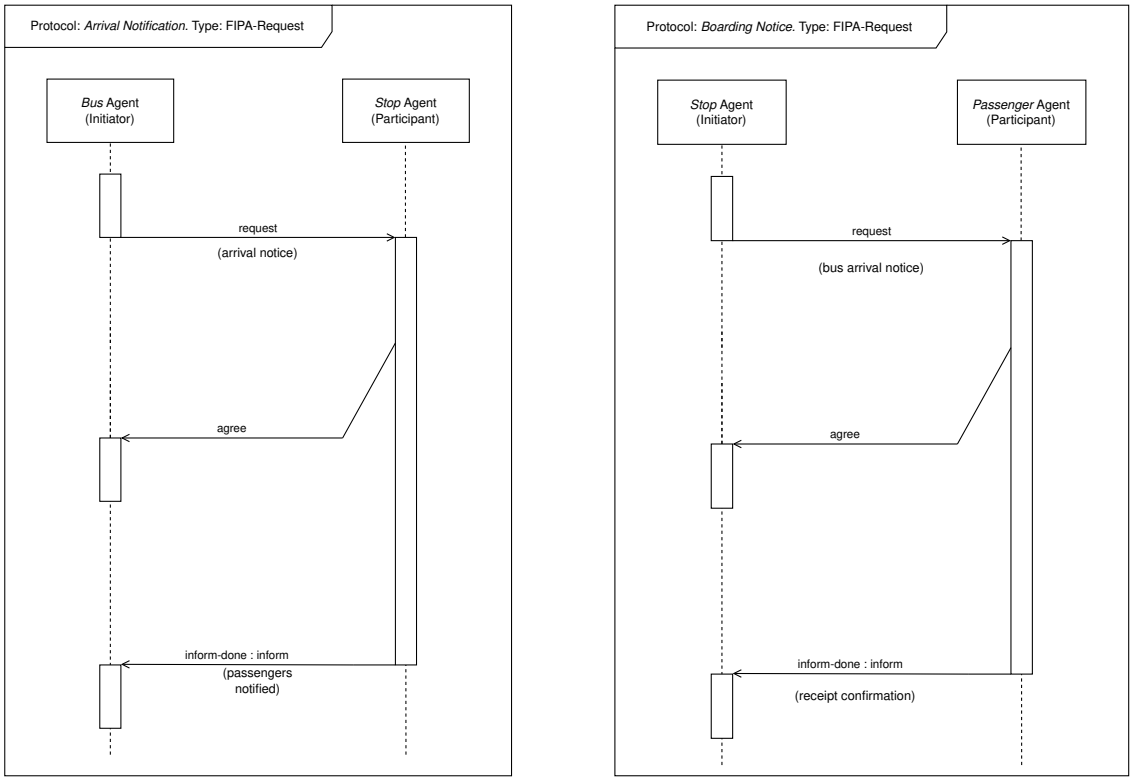
$$g_h(t) = \delta_h + \frac{2\,(\delta_{h+1} - \delta_h)}{T}\left(t - t_h - \frac{T}{2}\right), \qquad t \in [t_h,\, t_h + T), \tag{1}$$

where t_h is the start of hour h, T its duration, N the number of steps it contains and $\delta_h = \tau_h / N$. The mean of g_h over the hour is δ_h, so the number of passengers generated in the hour is still τ_h; only the instants at which they appear are redistributed.

Note that generating passengers during hour h requires τ_{h+1}, the number of passengers of the following hour. Each value is therefore sampled one hour in advance. At the start of the simulation both τ_5 and τ_6 are sampled, where 5 and 6 are its first two time slots, and at the start of every later hour h the value sampled an hour earlier becomes τ_h and a new τ_{h+1} is drawn.

In the first solution each generated passenger is an agent linked to its stop. In the second, generation increases an integer counter at the stop, and the instant of generation of each passenger is stored in a list, so that waiting times can still be obtained.

Selection of alighting passengers The selection of the passengers that alight must correspond to the fitted distributions. A destination cannot be assigned to a passenger when it is created, because the stop at which it alights would depend on which bus carries it and on the moment the bus reaches each stop; fixing it in advance would be arbitrary. Moreover, any such assignment would make the number of alightings per arrival an emergent quantity,



(a) Arrival notification (b) Boarding notice

Figure 2: FIPA-Request interaction protocols from first solution (2/3)

sensitive to delays, bunching and refusals for lack of capacity, and it would deviate from the alighting distributions, which are fitted per arrival.

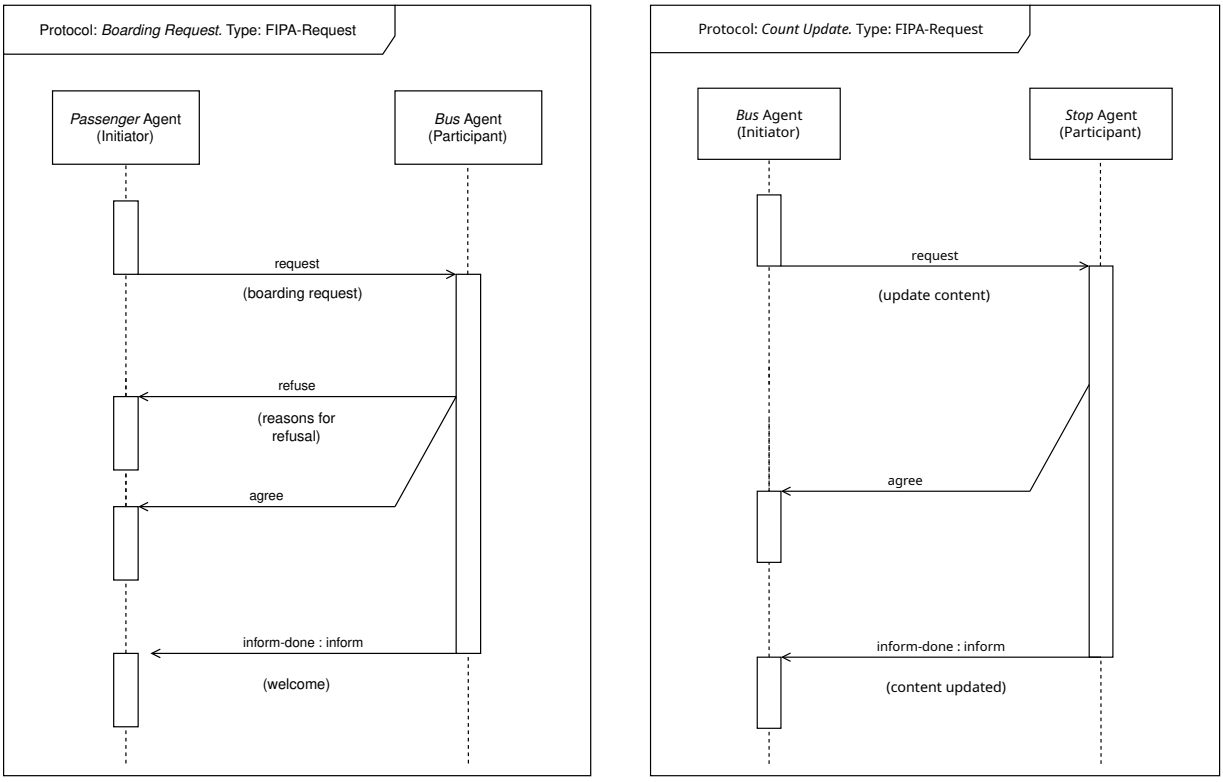
The selection is therefore made by the environment at the moment of alighting, the only point at which an empirical distribution supports it. In the first solution it is triggered by the bus when it starts the alighting query: a value is drawn from the alighting distribution of the next stop and time slot, and its integer part n gives the number of passengers to alight. The n passengers on board that waited longest at their stop are marked. The criterion is deterministic, and choosing it over any other is irrelevant to the purpose of the model, which is to reproduce the observed alighting counts. The mark is the interface between the global selection and the reactive behaviour of the passenger, which still carries out the corresponding protocol. At the last stop of each direction all passengers alight.

In the second solution the stop invokes the same mechanism when it receives the arrival query, and transmits the resulting number to the bus. In neither solution does the knowledge behind the selection belong to an agent, they only transmit its result.

Time at a stop In simulation steps, the exchange of messages at a stop takes the same time whatever the number of passengers involved. The time a bus remains at a stop is therefore extended to reflect the real duration of boarding and alighting, under two premises. First, boarding and alighting take place in parallel, one through each door, so the duration is bounded by the slower of the two flows and not by their sum. Second, each passenger takes an approximately constant time t_p. The time added at a stop is

$$E = \max\big(\max(b,\, a)\, t_p,\; E_{\min}\big), \tag{2}$$

where b and a are the numbers of boarding and alighting passengers and $E_{\min}$ is the time taken by the message exchanges, which is the minimum. As a limitation, this minimum also applies at stops with no activity, where a real bus would not stop; the bias is a small constant overhead per stop.



(a) Boarding request (b) Count update

Figure 3: FIPA-Request interaction protocols from first solution (3/3)

Table 2
Differences between the two solutions.

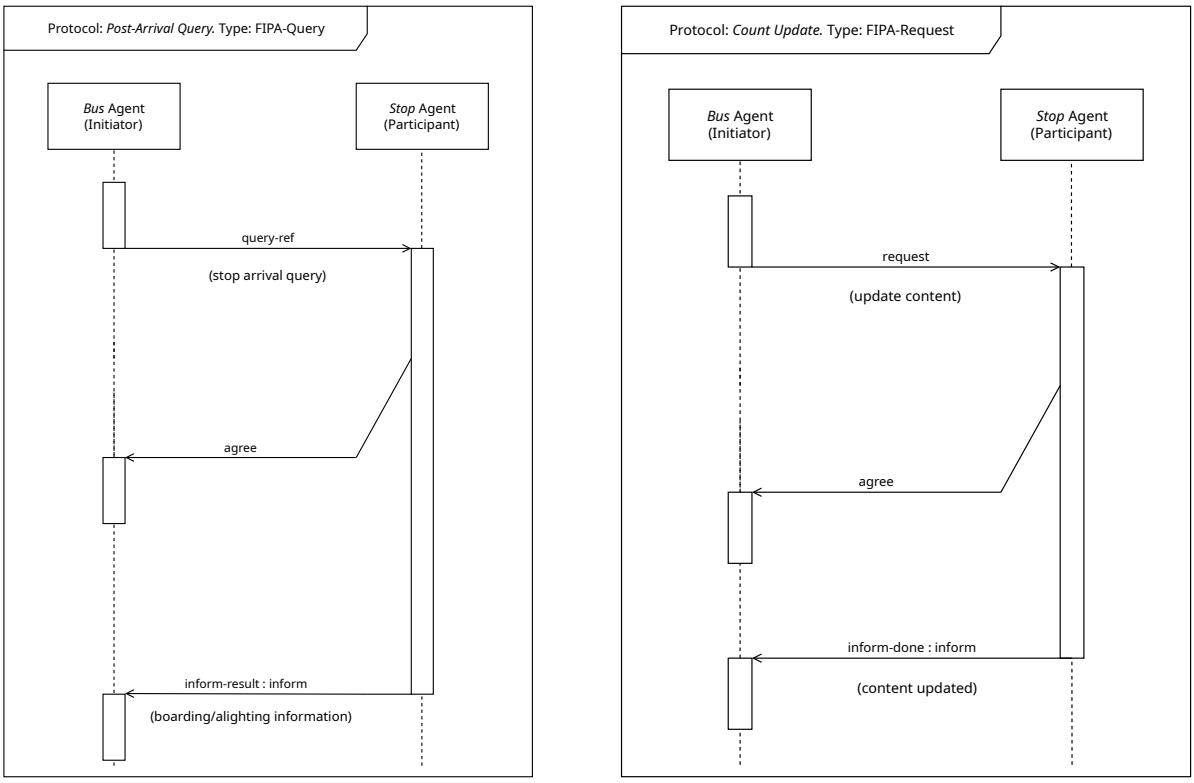
	First solution	Second solution
Passenger	Agent, from its creation at a stop until it alights	Not an agent; counted at stops and buses
Occupancy of a bus	List of the passenger agents on board	Number of passengers on board
State of a stop	List of the passenger agents waiting	Number of passengers waiting and list of their instants of generation
Role of a stop	Mediates between bus and passengers	Reports to the bus the passengers waiting and alighting
Alighting	Each marked passenger answers the bus	The stop transmits the number
Waiting time	Sent by each passenger in its boarding request	Obtained by the stop from the instants of generation
Protocols	Six	Two
Messages per stop visit	Grow with the passengers involved	Six

2.3. Differences between the two solutions

Both solutions share the bus and stop agents, the schedule and capacity of the buses, the movement between stops, the generation of passengers, the selection of alighting passengers, the time at a stop and the recorded outputs. Table 2 summarises where they differ.

2.4. Recorded outputs

Both solutions record the same data, so that the indicators are obtained in the same way. Each time a bus serves a stop, the following are stored: the passengers that boarded, the passengers that could not board because the bus was



(a) Arrival query (b) Count update (duplicated)

Figure 4: FIPA-Request interaction protocols from second solution

full, the passengers that alighted, the occupancy of the bus after the operation, and the travel time from the previous stop. The waiting time of each passenger that boarded is stored as well. Departures from the first stop that take place later than scheduled are recorded as delays.

The complete outputs are recomputed into defined Key Performance Indicators (KPIs) for demand absorption measurement.

This indicators can be divided into three main categories, depending of the scope they are aimed at: general values, route values, and trip values.

The general values presents the following indicators with its comptue method

3. Data calibration

3.1. Operational data

The simulator is built on the operational records of line E01 provided by EMT Madrid, covering an eight-month window starting July 2025. The records follow a two-table model linked by a trip identifier. Each departure from a terminal is an entry of the first table, with its date, time, time slot, direction, vehicle and type of day. Each arrival of that trip at a stop is an entry of the second table, with the time of arrival, its time slot, and the number of passengers that boarded and alighted. The occupancy after each stop is computed as the previous occupancy plus the boardings minus the alightings.

Cleaning focus on the boarding, alighting and occupancy fields, where negative values and extreme values derived from APC recording errors were found. The cleaning also involved the removal of any recorded occupancy value above a stablished threshold of 80 passengers. The removed records amount, on average, to one record every two days of data at each stop.

The study is restricted to normal days, defined as working days without free fare. Weekends and public holidays are excluded because the window starts in July 2025, so their number of observations is small compared with working

Table 3
Share of combinations whose fitted distribution is not rejected, per direction of the line.

Variable	Towards Cibeles	Towards La Peseta
Boardings	90.60%	87.56%
Alightings	93.61%	90.91%
Travel times	95.49%	90.43%
Occupancy (validation)	94.39%	88.60%

days, and parameters fitted on them would be biased by the small sample. Days with free fare are excluded because the outlier analysis showed that their occupancy is systematically higher than on the remaining working days, and they add noise to the distributions. The window does not cover a full annual cycle, so seasonal effects cannot be separated from the data, and the fitted parameters describe the observed period.

3.2. Calibration of the inputs

The simulator receives four inputs: the boardings and alightings at each stop, the travel times between consecutive stops, and the schedule of the buses. The first three are fitted per time slot, using the slot of the moment at which the bus serves the stop; the schedule is taken directly from the operator, as the list of departures of each physical bus.

Boardings and alightings are modelled separately, and no origin-destination matrix is estimated (Section 1). They are also modelled at different levels, according to how they are used. Boardings are aggregated per stop and hour, since the simulator uses them as the number of passengers that appear at the stop during that hour (Section 2.2). Alightings are modelled per bus arrival, since that is the moment at which the simulator draws them.

Unlike occupancy, boardings and alightings are not normal in all cases: many stop and slot combinations show values close to zero and long right tails. Three candidate distributions are therefore considered: normal, negative binomial and geometric. For each stop and time slot, outliers are first removed with the median absolute deviation, which does not assume any particular distribution [21]; a goodness-of-fit test is then run for each candidate, and the candidate with the lowest Akaike information criterion [22] among those not rejected is selected. When all three are rejected, the one with the lowest criterion is kept. Overall, the normal distribution is selected in 62.4% of the cases, the negative binomial in 23.9%, more often for boardings than for alightings, and the geometric in the remaining 13.7%.

Travel times between consecutive stops showed signs of normality from the first visual inspection, so they are fitted only to normal distributions, per segment and time slot. Table 3 gives the share of stop or segment and time slot combinations whose fitted distribution is not rejected.

Since alightings are drawn per bus arrival and independently of the number of passengers on board, the number of alightings over a period grows with the number of buses that serve the stop. The parameters are therefore tied to the service observed in the data, and the model reproduces that service; it is not suited to evaluating changes of frequency without recalibration.

3.3. Reference for validation

Occupancy is not an input of the simulator. It results from the combination of boardings, alightings and the number of buses. It is used after the simulation as the reference to validate the model. Its normality was studied per stop and time slot with a test based on skewness and kurtosis at a significance level of 0.05 [23, 24], with outliers detected by the generalised ESD test [25]. Treating all working days together gives a higher share of non-rejected combinations than treating each weekday separately, so no distinction by weekday is made in the rest of the work. For this variable, the time slot is that of the departure of the trip from its terminal. Rejections concentrate in the last time slot of the day, which has the fewest buses and therefore the fewest observations.

The terminal stops are excluded from this reference. It was reported that the transition between consecutive trips is not always recorded at the right moment, so the passengers boarding the next trip can be counted as occupancy at the last stop of the previous one, and vice versa.

3.4. Parametric implementation

The fitted parameters are loaded before the simulation starts and updated every hour of simulated time. Values are drawn with different methods for each family according to what the platform offers. Normal values use the native

generator of GAMA, based on the Box–Muller transform [26]. The negative binomial is sampled as a Gamma–Poisson mixture [27], which is valid for non-integer values of r:

$$\lambda \sim \text{Gamma}\left(r,\ \frac{1-p}{p}\right), \qquad X \mid \lambda \sim \text{Poisson}(\lambda), \tag{3}$$

with the Gamma distribution in shape and scale form. The geometric distribution has a closed-form inverse:

$$X = \left\lfloor \frac{\ln U}{\ln(1-p)} \right\rfloor, \qquad U \sim \mathcal{U}(0,1), \tag{4}$$

with support $\{0, 1, 2, \dots\}$, consistent with the parametrisation used in the fitting.

4. Implementations and results

Both solutions are implemented in GAMA [28], a platform dedicated to agent-based modelling and simulation. It was chosen for two of its features. Its FIPA skill provides each agent with a mailbox and supports the FIPA communicative acts and interaction protocols, as well as conversations whose sequence is defined by the modeller [29]. Its headless mode runs experiments from the command line without the graphical interface, so the replicas can be executed in batch [30].

4.1. Case study

For the simulation characterization, the layout of route Express 01 is used. The route operates between the central square *Cibeles*, located in the city center of the Spanish capital, and *La Peseta*, located in the southern neibourhood of Carabanchel. The route has 15 and 12 stops for each direction respectively, and its core behavour reflects the day by day of madrid's workers: Going to the center earlier in the morning to work and coming back home after lunch. The route's prime time in Cibeles's direction comes between 7 and 9 in the morning, boarding passengers in the first stops of the trip, and alighting them at the final ones. The same behaviour is inversed in La Peseta's direction, presenting its prime time after 3 pm, capturing the tendency of workers to return to its homeplace.

4.2. Simulation setup

Table 4 lists the parameters of the experiments. The time step Δt, the time per passenger t_p and the minimum time at a stop $E_{\min}$ are those of Eq. (2), and the capacity of the buses is the one adopted in Section 3.1.

Table 4
Simulation parameters.

Parameter	Value
Line	E01 (Cibeles–La Peseta)
Physical buses	12
Scheduled trips per day	163
Bus capacity	80 passengers
Start of the simulation	05:00
Time step (Δt)	2 s
Time per passenger (t_p)	5 s
Minimum time at a stop ($E_{\min}$)	$4\,\Delta t$
Replicas per solution	1000

4.3. Evaluation

5. Conclusions

CRediT authorship contribution statement

H. Álvarez: Aqui va la contribucion mia. **J. Carbó:** Aqui va la contribucion de Javier/José Manuel. **J.M. Molina:** Aqui va la otra contribucion de Javier/José Manuel.

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